

Alternative Price Zone Configurations for Germany

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ACER proposed 4 alternative configurations for Germany

- Identified based on locational marginal prices (BZR, 2022)
- Configurations should be based on long-term structural congestion in the transmission grid
- Goal: increase economic efficiency & cross-zonal trading opportunities

Analysis

- Time periods:
 - 1989, 1995, 2009
- Clustering features:
 - Latitude & longitude
 - Nodal prices
- Clustering algorithm:
 - Constrained K-Means

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	A	B	C	D	E	F	G	H	I	J	K	L
1	Climate year	Week	Month	Date	Hour	MTU index	Node ID	Market node	Country	Nodal price (EUR/MWh)		
2	1989	4	1	22.01.	0	1	_00197ca9ccb1	DE00	DE	50.8877		
3	1989	4	1	22.01.	2	3	_00197ca9ccb1	DE00	DE	50.8367		
4	1989	4	1	22.01.	4	5	_00197ca9ccb1	DE00	DE	51.3413		
5	1989	4	1	22.01.	6	7	_00197ca9ccb1	DE00	DE	53.6848		
6	1989	4	1	22.01.	8	9	_00197ca9ccb1	DE00	DE	58.3985		
7	1989	4	1	22.01.	10	11	_00197ca9ccb1	DE00	DE	58.2105		
8	1989	4	1	22.01.	12	13	_00197ca9ccb1	DE00	DE	58.0341		
9	1989	4	1	22.01.	14	15	_00197ca9ccb1	DE00	DE	61.1236		
10	1989	4	1	22.01.	16	17	_00197ca9ccb1	DE00	DE	61.4548		
11	1989	4	1	22.01.	18	19	_00197ca9ccb1	DE00	DE	61.442		

Analysis



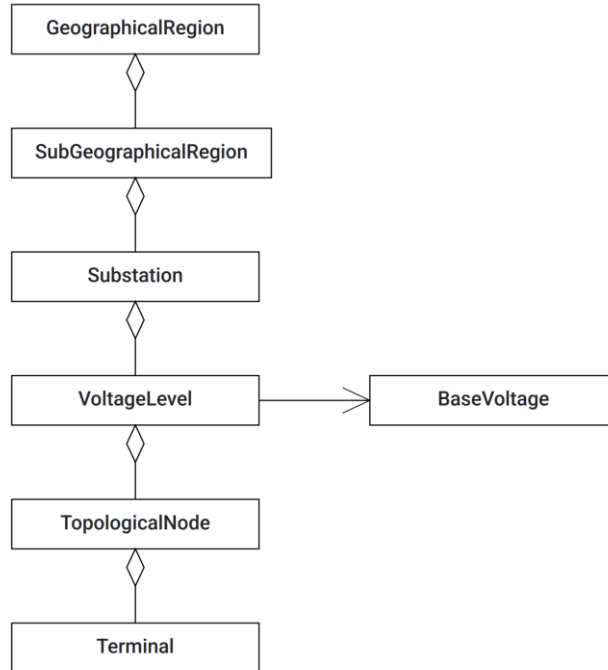
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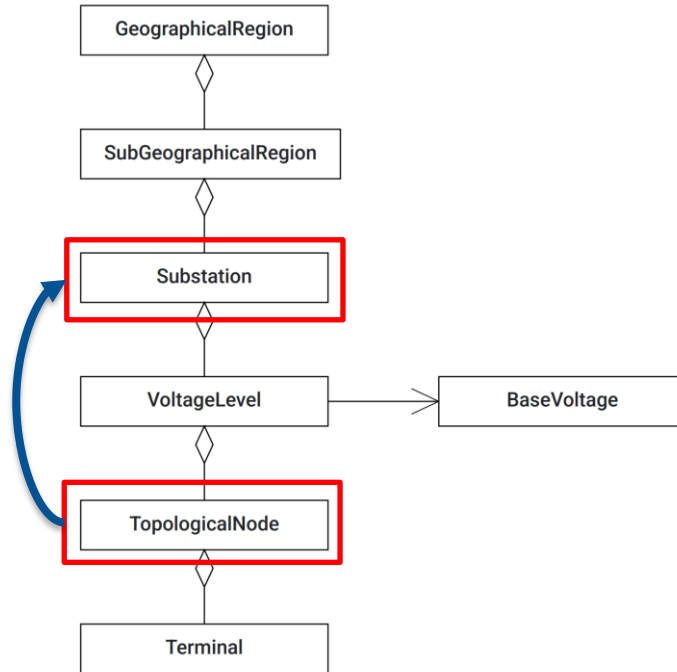
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Retrieving geographical coordinates of nodes



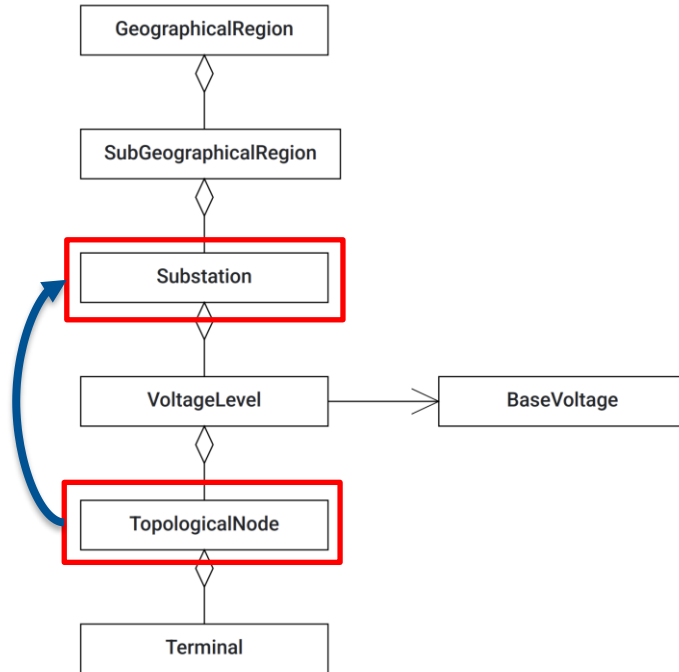
Credits to Nicolas Schmitt

Retrieving geographical coordinates of nodes



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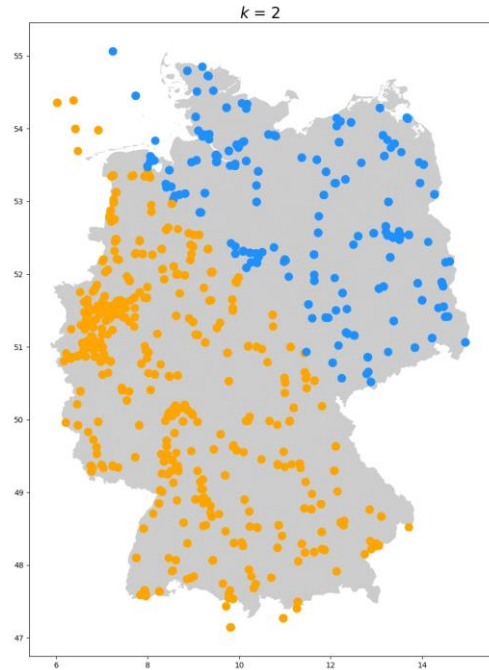
Retrieving geographical coordinates of nodes



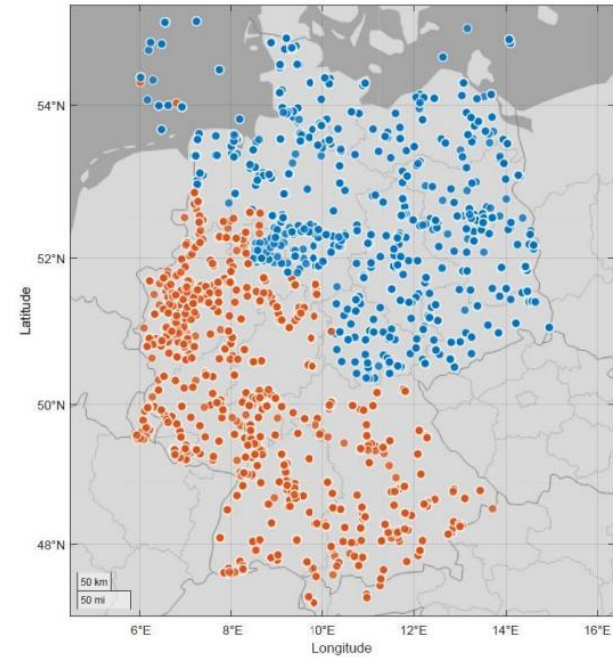
Credits to Nicolas Schmitt

K-Means based on **geographic coordinates**

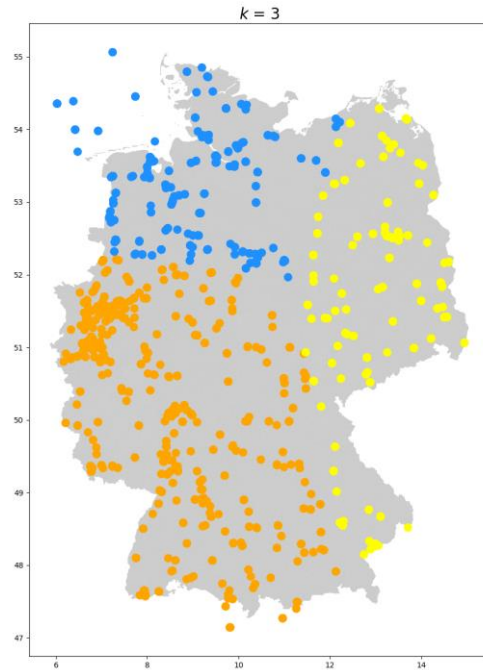
K-Means based on latitude & longitude



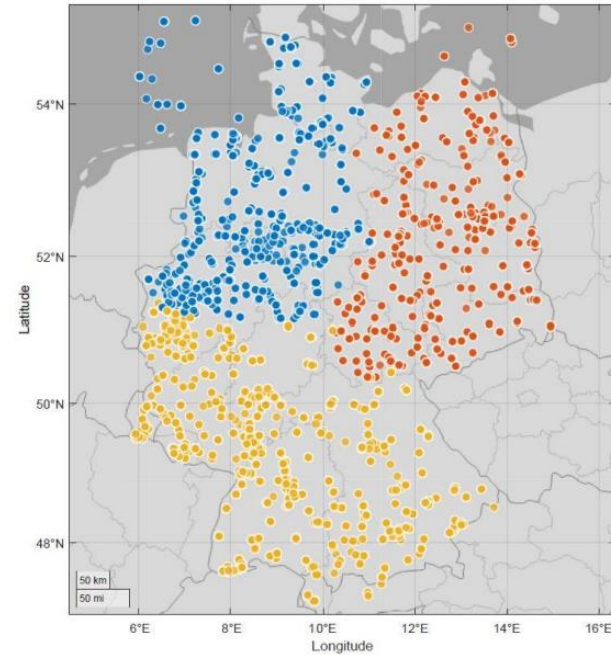
DE2 (spectral)



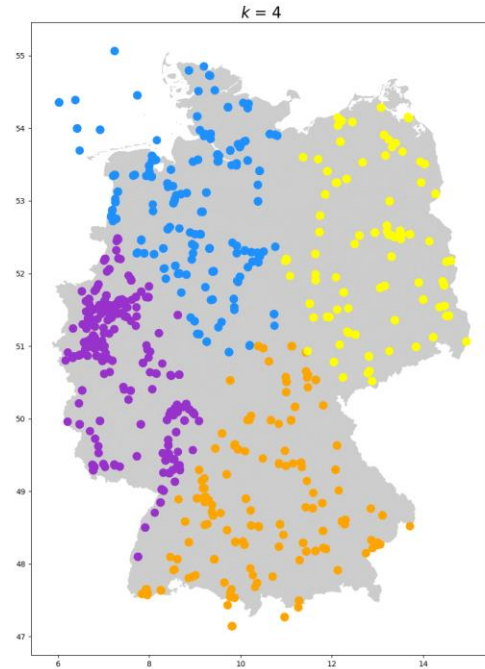
K-Means based on latitude & longitude



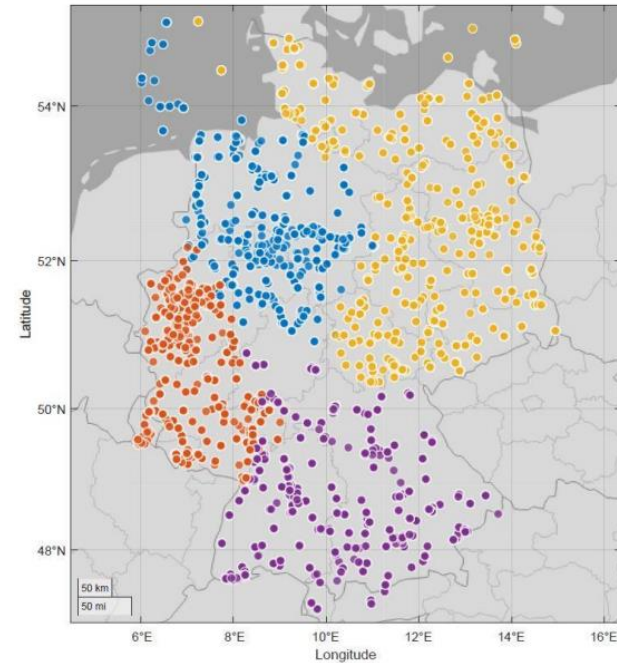
DE3 (spectral)



K-Means based on latitude & longitude

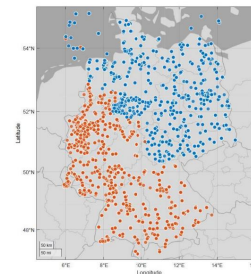
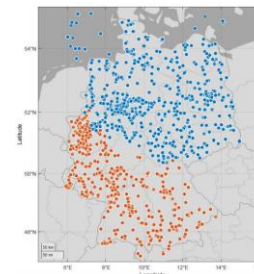


DE4 (spectral)

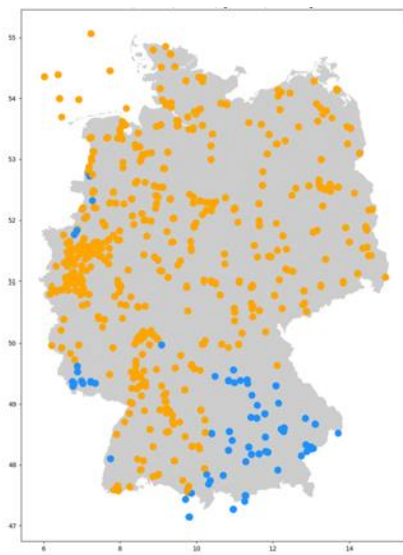


K-Means based on **prices**

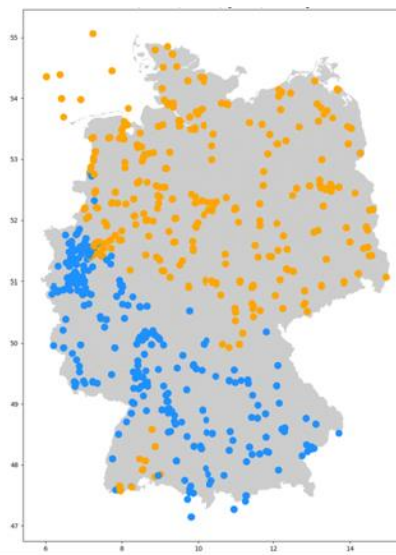
2 zones



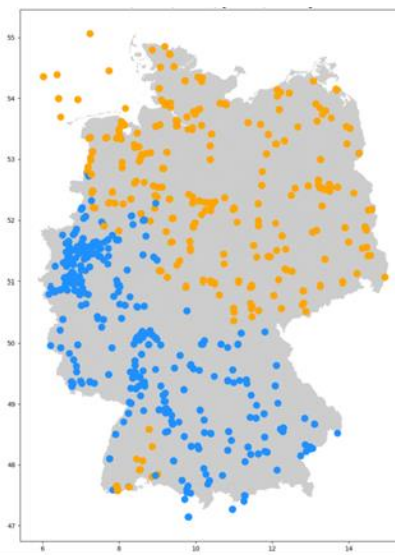
1989



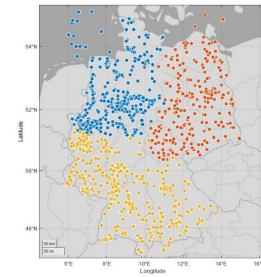
1995



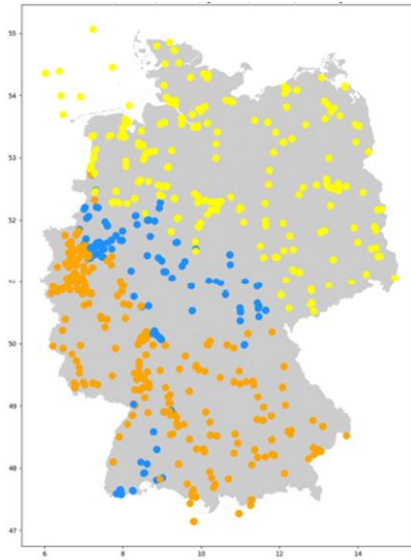
2009



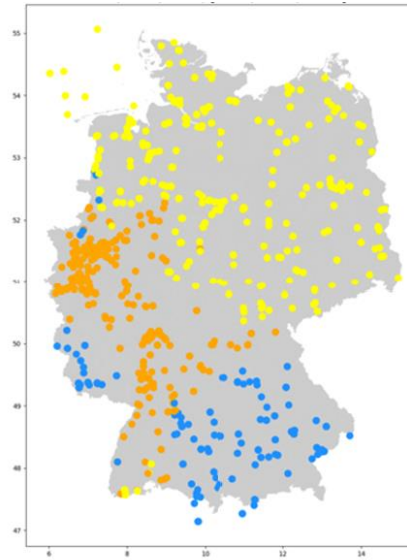
3 zones



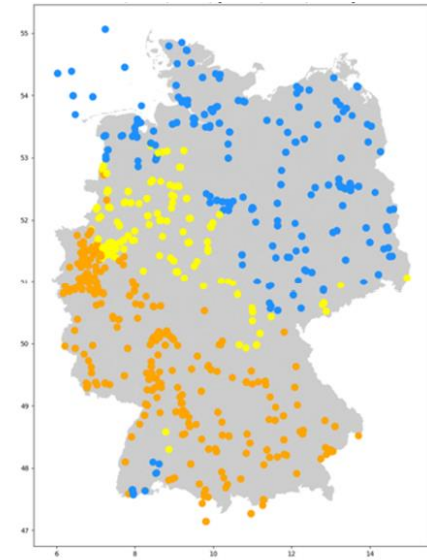
1989



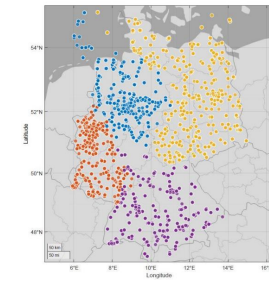
1995



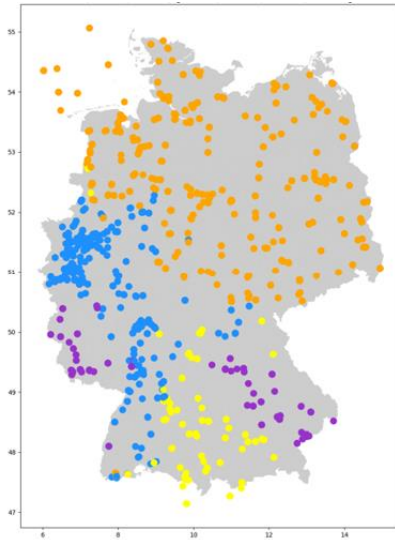
2009



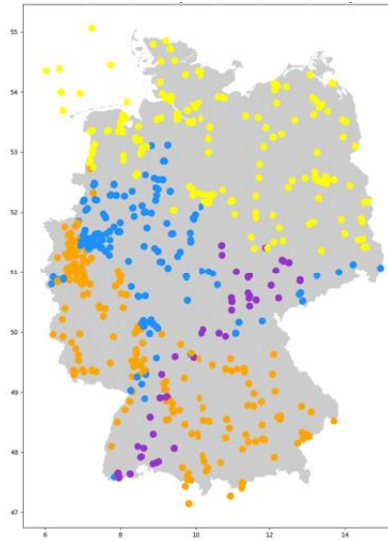
4 zones



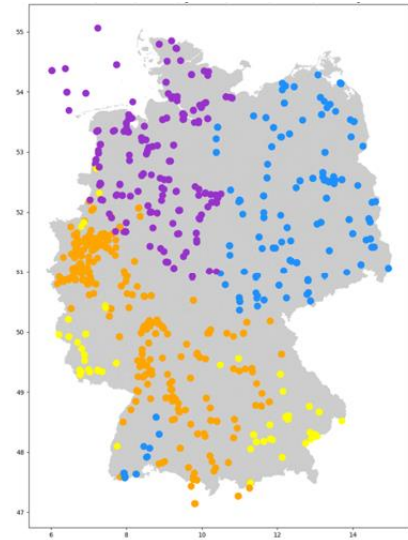
1989



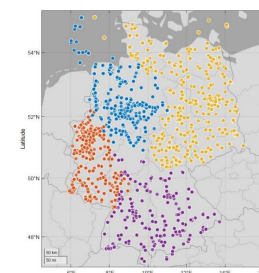
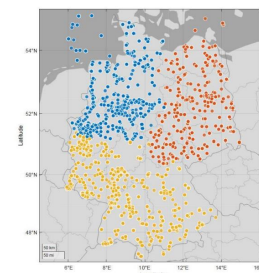
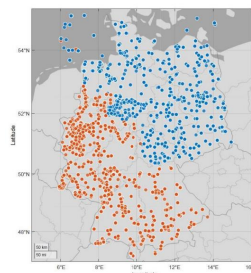
1995



2009

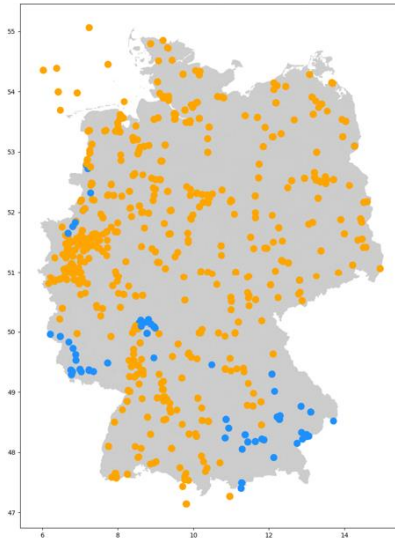


What if we consider the **prices** corresponding to a **single day**?

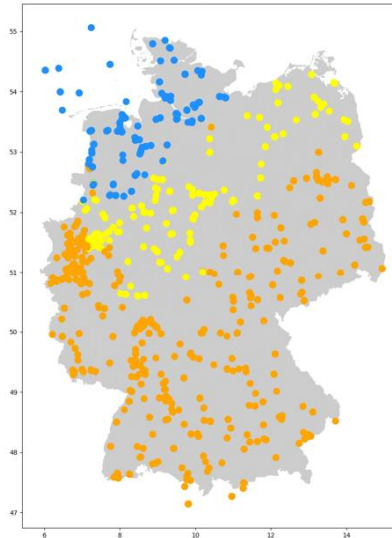


January 22, 1989

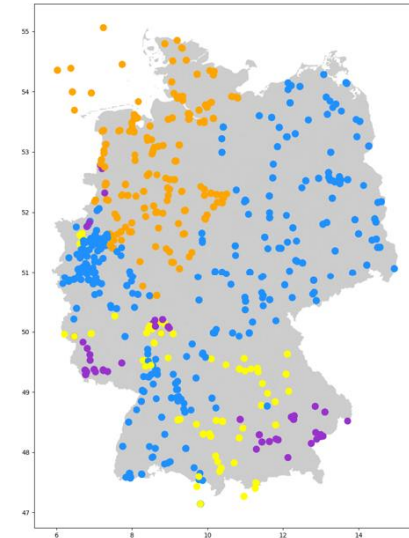
2 zones



3 zones



4 zones



Discussion

1. K-Means based on geographic coordinates → similar configurations
2. K-Means based on prices:
 - different period → different configuration

Thanks!